Let $(X, \mathbb{P})$ be a probability space and $F\colon X \times I \circlearrowleft$ a skew product defined by

$$F(x, y) = (g(x), f_x(y))$$

where $g\colon X \circlearrowleft$ is a measure-preserving map.

For a bounded function $\varphi\colon I \to \mathbb{R}$, we denote

$$U_\varphi(x, y) = \limsup_{n\to\infty} \frac{1}{n} \sum_{i=0}^{n-1} \varphi(\pi_2(F^i(x, y))).$$

Since φ is bounded, $U_\varphi(x, y) = U_\varphi(F(x, y))$ everywhere. Indeed, since for any pair of sequences $(r_n)_n, (s_n)_n \subseteq \mathbb{R}$ we have

$$|\limsup r_n - \limsup s_n| \leq \limsup |r_n - s_n|,$$

we conclude that

$$|U_\varphi(x, y) - U_\varphi(F(x, y))| \leq \limsup_{n\to\infty} \frac{1}{n} |\varphi(y) - \varphi(\pi_2(F^n(x, y)))| = 0.$$

For a fixed bounded φ, suppose the following is true:

Hypothesis 1. There exists a measurable function $\ell\colon I \to \mathbb{R}$ and a family $\{\Omega_y\}_{y\in I}$ of subsets of X, each with full measure $\mu(\Omega_y) = 1$, such that $U_\varphi(x, y) = \ell(y)$ for all $x \in \Omega_y$.

Even if U_φ is invariant for every point, it's not clear if $\ell(f_x(y)) = \ell(y)$ a.e. (let alone for every y) because we're dealing with the set $\{(x, y) : g(x) \in \Omega_{f_x(y)}\}$ (which presumably could have small measure), so the argument of [1] seems incomplete. If the pushforward of Lebesgue by the f_x are absolutely continuous to Lebesgue, however, we're good:

Lemma 1. *Suppose $f_x m \ll m$ for $\mathbb{P}$-a.e. $x \in X$. Then there exists a set $\Omega \subseteq X \times I$ with full measure $(\mathbb{P} \otimes m)(\Omega) = 1$ where $\ell(f_x(y)) = \ell(y)$ for all $(x, y) \in \Omega$.*

Proof. Let

$$\Omega = \{(x, y) : x \in \Omega_y \wedge g(x) \in \Omega_{f_x(y)}\}.$$

It's clear from the definition of the $\{\Omega_y\}_{y\in I}$, combined with the fact U_φ is invariant everywhere, that $\ell(f_x(y)) = \ell(y)$ in the set Ω. Also,

$$\begin{aligned}
(\mathbb{P} \otimes m)(\Omega^c) &\leq \underbrace{(\mathbb{P} \otimes m)(\{(x, y) : x \notin \Omega_y\})}_{0 \text{ by Fubini}} + (\mathbb{P} \otimes m)(\{(x, y) : g(x) \notin \Omega_{f_x(y)}\}) \\
&= \int_X m(\{y : g(x) \notin \Omega_{f_x(y)}\})\, d\mathbb{P}(x) \\
&= \int_X ((f_x)_* m)(\{y : g(x) \notin \Omega_y\})\, d\mathbb{P}(x).
\end{aligned}$$

Since $g_*\mathbb{P} = \mathbb{P}$, we have

$$(\mathbb{P} \times m)(\{(x, y) : g(x) \notin \Omega_y\}) = (\mathbb{P} \times m)(\{(x, y) : x \notin \Omega_y\}) = 0.$$

By Fubini, the function $x \mapsto m(\{y : g(x) \notin \Omega_y\})$ is $\mathbb{P}$-a.e. equal to zero. Then the hypothesis $(f_x)_* m \ll m$ implies that $x \mapsto (f_x)_* m(\{y : g(x) \notin \Omega_y\})$ is zero $\mathbb{P}$-a.e., so the integral above is zero. $\square$

Corollary 1. *If for $\mathbb{P}$-a.e. $x \in X$ there is some σ-finite invariant measure μ_x for f_x such that $m \sim \mu_x$, then a set Ω as in lemma 1 exists.*

Proof. By Radon-Nikodym, write $m = w\mu_x$. Then $(f_x)_* m = (w \circ f_x)\mu_x \ll m$. □

Now we will see the assumptions of the lemma are satisfied on the zero-Schwarzian context of [1], where each $f_x : I \circlearrowleft$ is an orientation-preserving diffeomorphism with zero Schwarzian derivative. Indeed, this condition means that there exists a measurable function $p : X \longrightarrow \mathbb{R}$ such that each $f_x = \gamma^{-1} \circ h_x \circ \gamma$ is conjugate to the translation $h_x(t) = t + p(x)$ via the change of coordinates

$$\gamma(y) = \log y - \log(1 - y).$$

Since the Lebesgue measure $m_{\mathbb{R}}$ on $\mathbb{R}$ is invariant for translations, the measure $\mu = \gamma_*^{-1} m_{\mathbb{R}}$ is invariant for every f_x. It's also σ-finite, because γ is a measurable bijection, and $m \sim \mu$ because γ^{-1} is a diffeomorphism onto $(0, 1)$ (thus locally Lipschitz).

The rest of the argument in [1] is still true if ℓ is invariant only in the full measure set Ω (instead of all (x, y) in a family of full-measure sets of $x \in X$, indexed by $y \in I$). Note that the full conclusion of the Theorem 6.2 is also still true (for all $y_0 \in (0, 1)$), because the function ℓ is monotone.

Bibliography

[1] Araceli Bonifant and John Milnor. “Schwarzian Derivatives and Cylinder Maps”. In: *American Mathematical Society* (2008), pp. 1–24.